

Algebra: Dividing Polynomials

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DIRECTIONS

Divide the polynomial using long division or synthetic division. Include remainder if any.

1 Divide using synthetic division:

$$(x^3 - 3x^2 + 4) \div (x + 1)$$

Answer: _____

2 Divide using long division:

$$(x^3 + 8) \div (x - 1)$$

Answer: _____

3 Divide using synthetic division:

$$(x^3 + 2x^2 - 4x - 8) \div (x - 2)$$

Answer: _____

4 Divide using long division:

$$(x^3 + x^2 - 2) \div (x - 1)$$

Answer: _____

5 Divide using synthetic division:

$$(x^3 - 5x^2 + 8x - 4) \div (x - 1)$$

Answer: _____

6 Divide using synthetic division:

$$(x^3 + 4x^2 + 5x + 2) \div (x + 2)$$

Answer: _____

7 Divide using long division:

$$(x^3 - 1) \div (x + 2)$$

Answer: _____

8 Divide using long division:

$$(x^3 - 3x + 2) \div (x - 3)$$

Answer: _____

9 Divide using synthetic division:

$$(x^3 + 2x^2 - x - 2) \div (x - 1)$$

Answer: _____

10 Divide using long division:

$$(x^3 - 2x^2 + 2x - 1) \div (x - 2)$$

Answer: _____



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Answer Key & Solutions

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TEACHER NOTES Synthetic division only works when the divisor is $(x-c)$. Bring down, multiply, add — repeat.

1 Divide using synthetic division:

$$= x^2 - 4x + 4$$

$$(x^3 - 3x^2 + 4) \div (x + 1)$$

2 Divide using long division:

$$= x^2 + x + 1 + \frac{9}{x-1}$$

$$(x^3 + 8) \div (x - 1)$$

3 Divide using synthetic division:

$$= x^2 + 4x + 4$$

$$(x^3 + 2x^2 - 4x - 8) \div (x - 2)$$

4 Divide using long division:

$$= x^2 + 2x + 2$$

$$(x^3 + x^2 - 2) \div (x - 1)$$

5 Divide using synthetic division:

$$= x^2 - 4x + 4$$

$$(x^3 - 5x^2 + 8x - 4) \div (x - 1)$$

6 Divide using synthetic division:

$$= x^2 + 2x + 1$$

$$(x^3 + 4x^2 + 5x + 2) \div (x + 2)$$

7 Divide using long division:

$$= x^2 - 2x + 4 + \frac{-9}{x+2}$$

$$(x^3 - 1) \div (x + 2)$$

8 Divide using long division:

$$= x^2 + 3x + 6 + \frac{20}{x-3}$$

$$(x^3 - 3x + 2) \div (x - 3)$$

9 Divide using synthetic division:

$$= x^2 + 3x + 2$$

$$(x^3 + 2x^2 - x - 2) \div (x - 1)$$

10 Divide using long division:

$$= x^2 + 2 + \frac{3}{x-2}$$

$$(x^3 - 2x^2 + 2x - 1) \div (x - 2)$$



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